

Day 04: Loops in Python

Learning Objectives

After completing this lesson, you will be able to:

Understand loops in Python

Learn different types of loops

Use for loop and while loop

Understand range() function

Use break, continue, and pass statements

Write loop-based programs

1. What are Loops?

Loops are used to execute a block of code repeatedly until a specific condition is satisfied.

Instead of writing the same code multiple times, loops help to reduce repetition.

Example:

Without Loop:

```
print("Python")
```

```
print("Python")
```

```
print("Python")
```

Using Loop:

```
for i in range(3):
```

```
    print("Python")
```

Output:

```
Python
```

```
Python
```

```
Python
```

2. Types of Loops in Python

Python provides two main types of loops:

for Loop

while Loop

3. for Loop

A for loop is used to iterate over a sequence such as list, string, or range.

Syntax:

```
for variable in sequence:
```

```
    statements
```

Example:

```
for i in range(5):
```

```
    print(i)
```

Output:

0

1

2

3

4

4. range() Function

The range() function generates a sequence of numbers.

Syntax:

```
range(start, stop, step)
```

Example:

```
for i in range(1,6):
```

```
    print(i)
```

Output:

1

2

3

4

5

Example with step:

```
for i in range(2,10,2):
```

```
    print(i)
```

Output:

2

4

6

8

5. Looping Through a List

Example:

```
fruits = ["Apple", "Banana", "Mango"]
```

```
for fruit in fruits:
```

```
    print(fruit)
```

Output:

Apple

Banana

Mango

6. while Loop

A while loop executes a block of code as long as the given condition is True.

Syntax:

```
while condition:
```

```
    statements
```

Example:

```
count = 1
```

```
while count <= 5:
```

```
    print(count)
```

```
    count += 1
```

Output:

1

2

3

4

5

7. Loop Control Statements

Python provides three loop control statements:

break

Used to stop the loop immediately.

Example:

```
for i in range(10):
```

```
    if i == 5:
```

```
        break
```

```
    print(i)
```

Output:

0

1

2

3

4

continue

Skips the current iteration and continues with the next iteration.

Example:

```
for i in range(5):
```

```
    if i == 2:
```

```
        continue
```

```
    print(i)
```

Output:

0

1

3

4

pass

Used as a placeholder when no action is required.

Example:

```
for i in range(5):
```

```
    pass
```

8. Nested Loops

A loop inside another loop is called a nested loop.

Example:

```
for i in range(3):
```

```
    for j in range(2):
```

```
        print(i,j)
```

Output:

0 0

0 1

1 0

1 1

2 0

2 1

9. Practice Programs

Program 1

Print numbers from 1 to 10

```
for i in range(1,11):
```

```
    print(i)
```

Program 2

Find Sum of Numbers

```
sum = 0
```

```
for i in range(1,6):
```

```
    sum = sum + i
```

```
print(sum)
```

Output:

15

Program 3

Multiplication Table

```
num = 5
```

```
for i in range(1,11):
```

```
    print(num*i)
```

Program 4

Print Even Numbers

```
for i in range(1,11):
```

```
    if i % 2 == 0:
```

```
        print(i)
```

Output:

2

4

6

8

10